URPS 2026

Investigating different pooling Strategies for Ridge Regression with Multiple Imputation

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April 2026

1 Introduction

Missing data is a common challenge in real-world datasets, especially in high-dimensional regression settings where multicollinearity is present. A widely used approach to handle missing data is multiple imputation by chained equations (MICE), which generates multiple completed datasets and combines the results using Rubin's rules, providing a flexible framework that accounts for statistical uncertainty across imputations. A key step in this framework is to pool the results across the imputed datasets. In practice, different pooling strategies can be applied in regression studies, such as pooling parameter estimates or pooling mean square errors, and these choices may lead to different conclusions.

In high-dimensional settings, penalized regression methods, such as ridge regression, are commonly used to address multicollinearity by introducing a penalty that shrinks coefficients and stabilizes the estimation. As a result, ridge regression introduces bias due to coefficient shrinkage. While standard multiple imputation frameworks provide valid inference under standard parametric models, their behavior with penalized regression remains unclear.

In this work, we evaluate how pooling strategies affect the accuracy and stability of estimating MSE as a target parameter relative to the complete-data benchmark. Specifically, We compare three pooling strategies: direct MSE pooling, coefficient mean pooling, and coefficient SVD-based pooling. The latter two belong to the class of coefficient pooling. We evaluate these strategies under two frameworks: a baseline framework and a train–test framework. The baseline setting provides a controlled environment in which the penalty parameter is held fixed across imputations, allowing us to isolate the effect of pooling strategies independently of tuning variability, while the train–test framework reflects practical out-of-sample evaluation. This setup allows us to better understand how the choice of pooling strategy affects MSE estimation in penalized regression settings with imputed data.

2 Methodology

2.1 Simulation Settings

Our primary objective is to evaluate how different pooling strategies affect the accuracy and stability of estimating MSE as a target parameter in practice, as measured by their deviation from the complete-data benchmark. To this end, we conduct a simulation study

examining how key data characteristics influence mean square error (MSE) estimation when ridge regression is combined with multiple imputation under different evaluation frameworks and penalty parameter selection strategies.

Specifically, we examine the effects of sample size, missingness proportion, collinearity, coefficient patterns and missing data mechanisms, because these factors play a central role in the estimation of MSE under incomplete data. Sample size influences estimation accuracy and variability, missingness proportion determines the severity of information loss, collinearity affects the stability of coefficient estimation, and the missing data mechanism determines the dependence between missingness and observed variables.

The sample size n is set with two levels, $n \in \{500, 1000\}$. The missingness proportion π is set with three levels, $\pi \in \{0.2, 0.5, 0.8\}$ representing low, moderate, and high levels of missingness. The collinearity parameter ρ is set with three levels $\rho \in \{0.2, 0.5, 0.8\}$, corresponding to weak, moderate, and strong correlation among predictors. To capture different structural settings, we consider two coefficient patterns (sparse and dense) and two missing data mechanisms (MCAR and MAR), resulting in four primary scenarios: sparse–MCAR, sparse–MAR, dense–MCAR, and dense–MAR.

Rather than adopting a full factorial design, we vary one primary factor at a time within each scenario while holding other factors fixed. This approach facilitates clearer interpretation of the effects of individual factors. To ensure comparability across settings, the random seed is held constant, and each scenario is repeated $R = 200$ times.

2.2 Data Generation

For each replicate, we first generate a complete dataset. Let $X \in \mathbb{R}^{n \times p}$ denote the predictor matrix, where the number of predictors is fixed at $p = 50$. This matrix X is generated as $X \sim \mathcal{N}(0, \Sigma)$, where the covariance matrix Σ follows a block autoregressive (AR) structure. Specifically, the predictors are divided into $K = 5$ independent blocks of equal size $b = 10$. Within each block, the covariance between two predictors is given by $\Sigma_{ij} = \rho^{|i-j|}$, while predictors across different blocks are assumed to be independent.

To reflect different signal configurations, we specify the true regression coefficients $\beta \in \mathbb{R}^p$ under either a sparse or dense structure. In the sparse setting, only the first $s = 10$ predictors have nonzero coefficients with randomly assigned signs and fixed magnitude signal = 1.0, while the remaining coefficients are set to zero. In the dense setting, all coefficients are

nonzero and are set as $\beta_j = \frac{\text{signal}}{\sqrt{p}}$ for $j = 1, \ldots, p$, ensuring comparable overall signal strength across settings. Given the predictor matrix X and coefficients β, the response is generated as $y = X\beta + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, 1)$ is independent of X.

2.3 Missing Mechanisms

Missingness is imposed on a predefined subset of predictors indexed by $M_{\text{risk}} = \{11, \ldots, 30\}$, corresponding to predictors X_{11} to X_{30}, while the remaining predictors are fully observed. Missing values are generated under either a Missing Completely at Random (MCAR) or Missing at Random (MAR) mechanism.

Under MCAR, each entry of the predictors in M_{risk} is independently set to missing with π. Under MAR, the probability of being observed follows a logistic model $\mathbb{P}(\text{observed}) = \text{logit}^{-1}\big(0.5X_1 + 0.5X_2 + 0.5y\big)$, where X_1 and X_2 are fully observed predictors. The coefficients are fixed at 0.5 to induce a moderate dependence of missingness on observed variables. The overall missingness proportion is calibrated to match the target level π, and entries are set to missing accordingly. This results in an incomplete dataset for each replicate.

2.4 Multiple Imputation

Multiple imputation is then applied on the incomplete data using the `mice` package with predictive mean matching, producing $m = 5$ completed datasets to account for uncertainty due to missingness.

2.5 Ridge Regression

Ridge regression estimates coefficients by solving $\hat{\beta}_{\text{ridge}} = \arg\min_\beta \{\|y - X\beta\|^2 + \lambda\|\beta\|^2\}$, where λ controls the strength of shrinkage. In our implementation, ridge regression models are fitted using the `glmnet` package in R. The penalty parameter λ is either fixed or selected via cross-validation, depending on the evaluation framework. For each dataset (complete or imputed), a ridge regression model is fitted accordingly.

2.6 Pooling Strategies

To combine results across the m imputed datasets, we compare three pooling strategies: direct MSE pooling, coefficient mean pooling, and coefficient SVD-based pooling. The latter two belong to the class of coefficient pooling.

MSE pooling. In this approach, a ridge regression model is fitted separately on each imputed dataset, and the corresponding mean squared error (MSE; see **Section 2.7**) is computed. Let $\text{MSE}^{(j)}$ denote the MSE from the j-th imputed dataset for $j = 1, \ldots, m$. The pooled MSE is then defined as the average across imputations:

$$\text{MSE}_{\text{pooled}} = \frac{1}{m} \sum_{j=1}^{m} \text{MSE}^{(j)}.$$

Coefficient pooling. In this approach, regression coefficients are first combined across imputations, then MSE is evaluated using the pooled coefficients. Specifically, we consider two strategies: mean pooling and SVD-based pooling. MSE is computed on each imputed dataset using the pooled coefficients and finally averaged across imputations.

Mean pooling. Let $\beta^{(j)} \in \mathbb{R}^p$ denote the estimated coefficient vector from the j-th imputed dataset. The pooled coefficient vector is defined as

$$\hat{\beta}_{\text{mean}} = \frac{1}{m} \sum_{j=1}^{m} \hat{\beta}^{(j)}$$

SVD-based pooling. To capture the dominant shared pattern across imputations, we construct a pooled coefficient vector using a low-rank representation of the coefficient matrix. Let

$$B = [\hat{\beta}^{(1)}, \ldots, \hat{\beta}^{(m)}] \in \mathbb{R}^{p \times m}$$

denote the matrix of estimated coefficients across imputations. We compute the singular value decomposition

$$B = UDV^{\top},$$

where V contains the right singular vectors. Let $v_1 \in \mathbb{R}^m$ denote the leading right singular vector. We define the signed normalized weight vector w by

$$w_j = \frac{v_{1j}}{\sum_{k=1}^{m} |v_{1k}|}, \qquad j = 1, \ldots, m.$$

The pooled coefficient vector is then given by

$$\hat{\beta}_{\text{svd}} = Bw.$$

To ensure interpretability, we align the sign of the pooled coefficients with the mean coefficient vector by flipping the sign of $\hat{\beta}_{\text{svd}}$ if $\langle \hat{\beta}_{\text{svd}}, \hat{\beta}_{\text{mean}} \rangle < 0$.

2.7 Evaluation Framework and Metrics

Baseline framework. In the baseline setting, we use a fixed penalty parameter λ to isolate the effect of pooling strategies from the variability due to tuning. We consider two fixed-λ strategies: (i) $\lambda = 1.0$, and (ii) λ selected by cross-validation on the complete data and then held fixed for all imputed datasets.

For each choice of λ, we first construct a complete-data benchmark by fitting ridge regression on the full data and computing

$$\text{MSE}_{\text{full}} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - x_i^\top \hat{\beta}_{\text{full}} \right)^2 .$$

Using the same fixed λ, ridge regression models are then fitted on each imputed dataset. For the j-th imputed dataset, the MSE is computed as

$$\text{MSE}^{(j)} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - x_{ij}^\top \hat{\beta}^{(j)} \right)^2 ,$$

where x_{ij} denotes the predictor vector from the j-th imputed dataset. Pooling is then applied as described in Section 2.6.

Train–test framework. In the train–test setting, each dataset is randomly split into training (80%) and test sets (20%). Missingness is imposed only on the training data, and multiple imputation is applied to generate m imputed training datasets. We first construct a complete-data benchmark by selecting the penalty parameter via cross-validation on the

complete training data, fitting a ridge regression model, and evaluating its performance on the test set:

$$\mathrm{MSE}_{\mathrm{full,test}} = \frac{1}{n_{\mathrm{test}}} \sum_{i=1}^{n_{\mathrm{test}}} \left(y_i - x_i^\top \hat{\beta}_{\mathrm{full}} \right)^2 .$$

For each imputed dataset j, the penalty parameter λ_j is selected via cross-validation on the corresponding imputed training data, which avoids the issue of double dipping, and a ridge regression model is fitted. All fitted models and pooled estimators are then evaluated on the same test set:

$$\mathrm{MSE}_{\mathrm{test}}^{(j)} = \frac{1}{n_{\mathrm{test}}} \sum_{i=1}^{n_{\mathrm{test}}} \left(y_i - x_i^\top \hat{\beta}^{(j)} \right)^2 .$$

Pooling is applied as described in Section 2.6.

Evaluation metric. In both frameworks, mean squared error (MSE) is used as the primary evaluation metric. In the baseline framework, MSE serves as an estimator of the target parameter, while in the train–test framework it evaluates out-of-sample performance.

To compare methods across simulation settings, we report the relative MSE ratio

$$R = \frac{\mathrm{MSE}_{\mathrm{method}}}{\mathrm{MSE}_{\mathrm{full}}},$$

where $\mathrm{MSE}_{\mathrm{method}}$ denotes the pooled MSE obtained from the imputed datasets under each pooling strategy as described in Section 2.6.

In the train–test framework, the corresponding ratio is defined as

$$R_{\mathrm{test}} = \frac{\mathrm{MSE}_{\mathrm{method,test}}}{\mathrm{MSE}_{\mathrm{full,test}}},$$

where $\mathrm{MSE}_{\mathrm{method,test}}$ denotes the corresponding test MSE under each pooling strategy as described in Section 2.6.

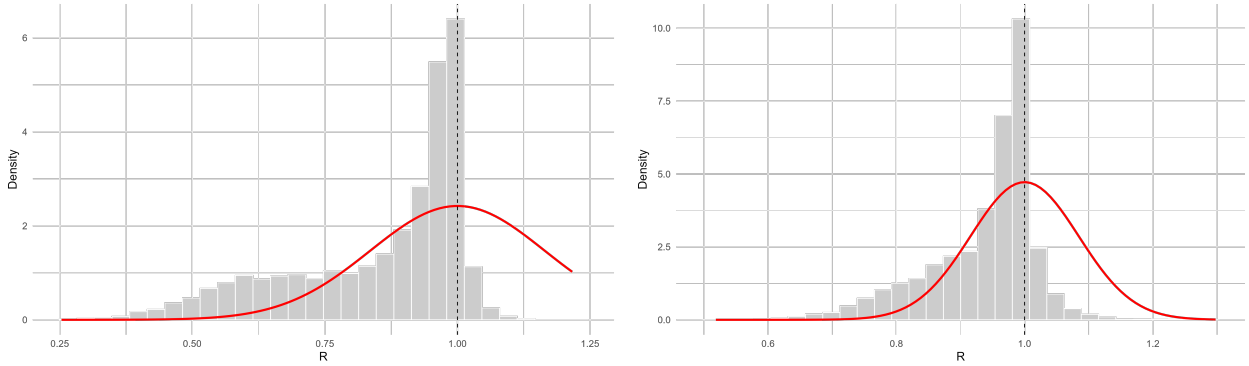
Values of R (or R_{test}) close to 1 indicates corresponding pooling strategy closely approximates the complete-data benchmark, while values farther from 1 indicate greater deviation. Across Monte Carlo replicates, we summarize the mean and standard deviation of R (and R_{test}).

3 Results

3.1 Effect of Lambda Selection on Direct MSE Pooling

We begin by comparing different λ selection strategies in the baseline framework and their impact on direct MSE pooling strategy. This is important because variability of λ can affect the stability of MSE estimates across imputations. We focus on direct MSE pooling strategy to isolate this effect.

As shown in **Figure 1**, when λ is selected by cross validation (see **Section 2.7**), the distribution of the ratio R is more dispersed and left-skewed with a longer tail toward smaller values, indicating higher variability. In contrast, fixing $\lambda = 1.0$ yields a more concentrated and more symmetric distribution around $R = 1$, indicating improved stability. Overall, cross validation introduces additional variability in R, while fixing λ leads to more stable but potentially less flexible estimates. This motivates us to use cross validation to select λ in subsequent analyses.

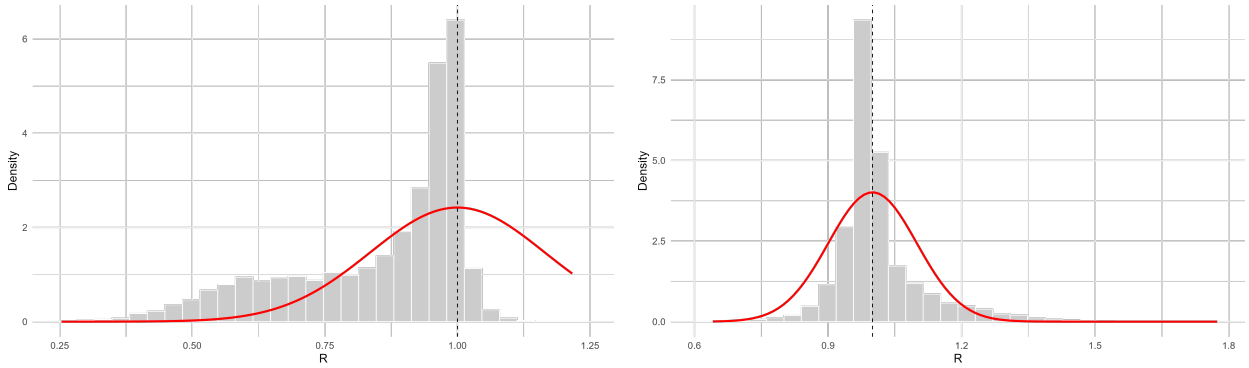


(a) CV-selected λ (b) Fixed $\lambda = 1.0$

Figure 1: Distribution of the relative MSE ratio R across Monte Carlo replicates under different lambda selection strategies. The dashed line indicates the ideal value $R = 1$, and the red curve shows a normal density with mean 1 and empirical standard deviation of R.

3.2 Coefficient Pooling Improves Stability

We next compare direct MSE pooling with coefficient pooling strategy under the cross validation-selected λ. As shown in **Figure 2**, coefficient pooling strategy produces the distribution of R that is more tightly concentrated around R = 1 compared to direct MSE pooling strategy, indicating improved stability. We'll further examine this pattern across different settings in **Figure 4**.



(a) Direct MSE pooling

(b) Coefficient pooling (mean)

Figure 2: Distribution of the relative MSE ratio R under CV-selected λ across Monte Carlo replicates for different pooling strategies. The vertical dashed line indicates the ideal value $R = 1$, and the red curve shows a normal density with mean 1 and empirical standard deviation of R.

3.3 Effect of Missingness on Estimation Stability

We further investigate the effect of missingness proportion on estimation performance under direct MSE pooling with a fixed CV-selected λ. As shown in **Figure 3**, at the low missingness ($\pi = 0.2$), the distribution of R is tightly concentrated around 1. As π increases to 0.5 and 0.8, the distribution becomes more dispersed and deviates further from 1. Overall, higher missingness leads to greater deviation from $R = 1$ and increased variability across all settings.

Notably, dense configurations produce estimates closer to $R = 1$, indicating more stable estimation of the target MSE. Other factors, such as missingness mechanism and sample size, have only minor effects, specifically with slightly higher variability under MAR and improved stability for larger samples.

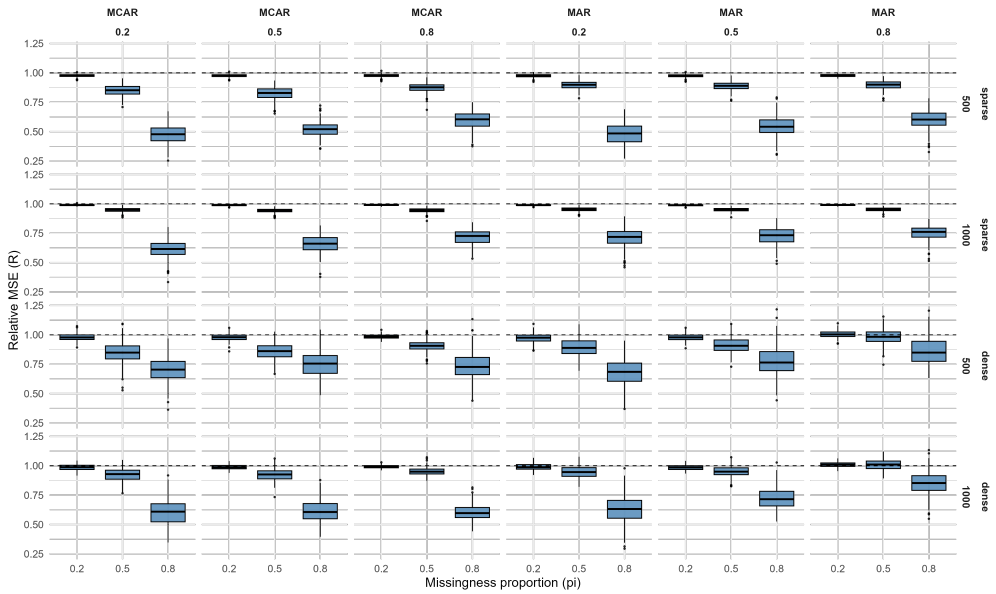


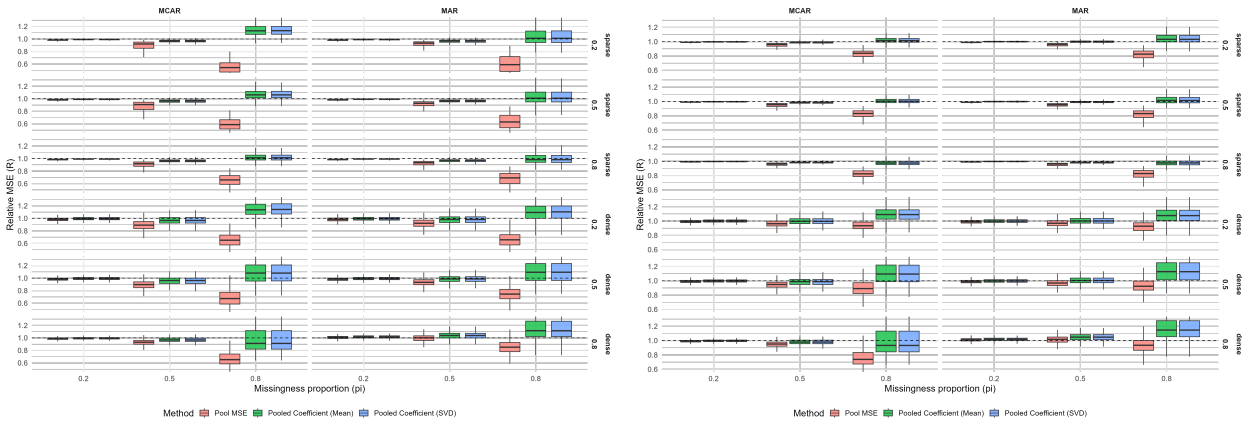
Figure 3: Effect of missingness proportion on the relative MSE ratio R under direct MSE pooling strategy. Results are shown across missingness mechanisms (MCAR vs. MAR), sample sizes ($n = 500, 1000$), and coefficient structures (sparse vs. dense). The dashed line indicates $R = 1$.

3.4 Comparison of Pooling Strategies

Across both lambda selection strategies, coefficient pooling strategies (mean and SVD) consistently produce estimates closer to the complete-data benchmark (R = 1) and exhibit lower variability than direct MSE pooling strategy. The improvement is pronounced at higher levels of missingness, where MSE pooling shows substantial deviation from the benchmark.

Notably, CV-selected λ generally yields estimates that are slightly closer to $R = 1$ and more stable, particularly under higher levels of missingness. In contrast, using a fixed $\lambda = 1$ leads to greater dispersion and larger deviations from the benchmark in some settings.

Mean and SVD coefficient pooling exhibit similar performance across most settings, with only minor differences observed. These patterns are consistent across missingness mechanisms, sample sizes, and coefficient structures **(Figure 4)**.

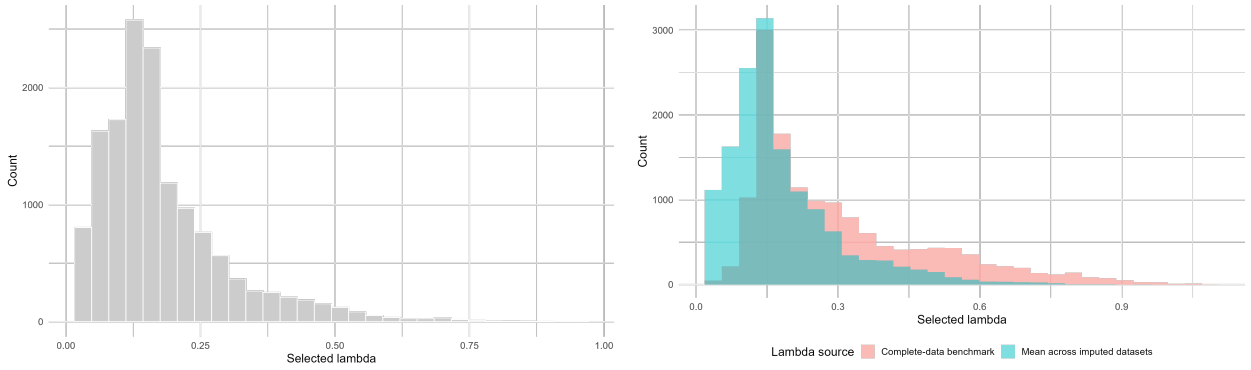


(a) CV-selected λ (b) Fixed $\lambda = 1$

Figure 4: Distribution of R across pooling strategies under different settings. Red: MSE pooling; Green: coefficient pooling (mean); Blue: coefficient pooling (SVD). The dashed line marks $R = 1$.

3.5 Out-of-Sample Performance and Robustness

We first examine the behavior of the CV-selected λ across imputations. The left panel shows a distribution concentrated at relatively small values with low variability, indicating stable tuning across imputations. The right panel compares the distribution of the CV-selected λ based on the complete training data and the average λ across imputed datasets. While the two distributions exhibit a broadly similar shape, the imputed-data distribution is noticeably shifted toward smaller values, suggesting that cross-validation tends to select smaller values of λ after imputation. This further implies that missingness also affects model selection, potentially influencing the comparisons across pooling strategies.



(a) λ across imputed datasets (b) comparison with complete data λ

Figure 5: Distribution of CV selected λ in the train–test framework.

As shown in **Figure 6**, direct MSE pooling exhibits substantially larger variability and more frequent extreme values, particularly at higher missingness levels ($\pi = 0.8$). In contrast,

coefficient pooling strategies produce estimates that are more concentrated and closer to the benchmark $R_{\text{test}} = 1$, indicating improved out-of-sample performance.

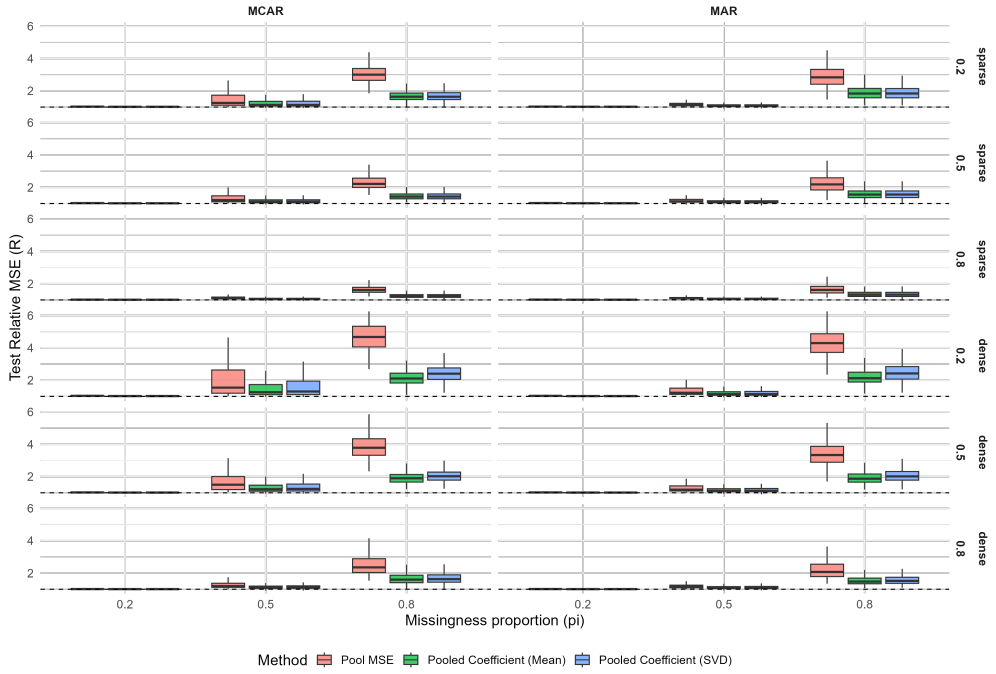
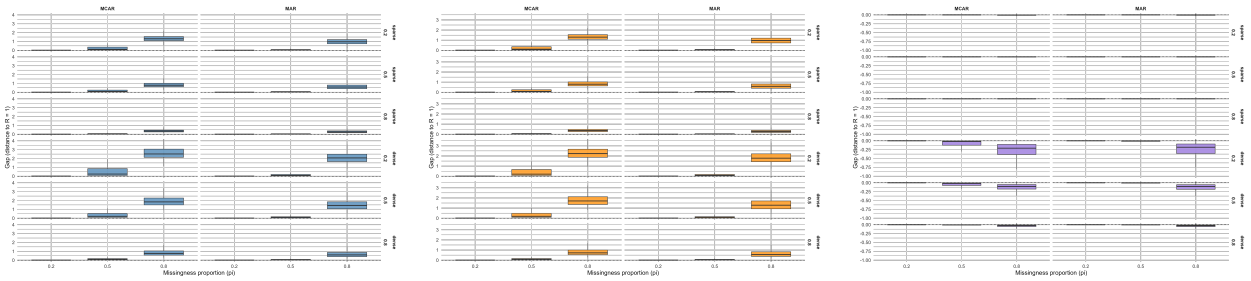


Figure 6: Distribution of R_{test} across pooling strategies. Red: MSE pooling; Green: coefficient pooling (mean); Blue: coefficient pooling (SVD). The dashed line marks $R_{\text{test}} = 1$.

These findings are further supported by gap-based comparisons (Figure 7), which show consistent advantages of coefficient pooling over direct MSE pooling strategy. Differences between mean and SVD pooling are generally small. In some settings, mean pooling yields slightly smaller deviation from the benchmark than SVD, although the overall performance of the two methods remains similar.



(a) MSE vs Mean (b) MSE vs SVD (c) Mean vs SVD

Figure 7: Gap metric comparison on test data across pooling strategies. The gap is defined as $\text{gap}(A, B) = |R_A - 1| - |R_B - 1|$, where positive values indicate that pooling strategy B yields a relative MSE closer to the benchmark than strategy A.

4 Conclusion

Overall, the simulation results demonstrate that both pooling strategy and penalty parameter selection substantially affect the accuracy and stability of the relative MSE ratio R. Direct MSE pooling exhibits high variability despite being centered near the benchmark. Using cross validation to select λ reduces variability compared to fixed λ, although it introduces additional dispersion due to tuning uncertainty. Missingness also affects the selection of the tuning parameter, with cross-validation tending to select smaller values of λ after imputation, highlighting the interaction between missing data handling and model selection.

Coefficient-based pooling methods consistently improve performance. Both mean and SVD coefficient pooling strategy produce estimates that are more tightly concentrated around $R = 1$, indicating improved stability and robustness across simulation settings. Differences between mean and SVD pooling are generally small; however, mean pooling often yields slightly smaller deviations from the benchmark, particularly in dense settings. Across all methods, increasing missingness proportion (π) leads to larger deviation from $R = 1$ and increased variability. Dense settings tend to yield more stable estimation of target MSE.

In the out-of-sample setting, these patterns remain consistent. Direct MSE pooling is the least stable and most sensitive to missingness, while coefficient pooling methods provide more reliable generalization performance. Increasing predictor correlation (ρ) improves stability for all methods, reducing variability and narrowing differences between pooling strategies.

Overall, coefficient pooling methods provide a more stable and reliable approach for estimating MSE under missing data, particularly in settings with moderate to high missingness.

5 Discussion

Why This Matters. This work addresses an important gap at the intersection of multiple imputation and penalized regression. While MICE is widely used to handle missing data and ridge regression is a standard regularization tool, there has been limited systematic evaluation of how imputation interacts with model estimation. Our findings demonstrate that the choice of pooling strategy is not merely a technical detail but has a direct impact on estimator stability and reliability. Particularly, aggregating performance metrics such as MSE across imputations can obscure variability introduced by the imputation process, whereas pooling at the coefficient level leads to more stable and interpretable results.

Practical Implications. From the perspective of application, these results suggest that practitioners working with multiple imputed datasets should prioritize coefficient-based pooling rather than direct pool MSE. This is particularly important when model tuning is involved, as variability in the tuning parameter (e.g., λ selected via cross-validation) can further amplify instability in downstream evaluation. This research provides a reproducible approach for evaluating imputation–model pipelines and highlights the importance of aligning pooling strategies with the structure of the underlying estimation problem.

Limitations. Several limitations should be noted. First, the simulation design considers a limited set of missingness levels and correlation structures, which may not fully capture the complexity of real-world data. Second, the analysis focuses exclusively on ridge regression, chosen for its convexity and computational stability. This excludes other widely used methods such as Lasso and elastic net, whose non-smooth penalties and variable selection behavior may interact differently with imputation.

Future Directions. Future work can extend this framework in several directions. One natural extension is to study sparsity-inducing methods such as Lasso and elastic net, particularly in high-dimensional settings. Another promising direction is to examine how imputation interacts with more complex models, such as neural networks, where tuning and optimization introduce additional variability. Understanding how imputation-induced uncertainty propagates through these models, and whether pooling strategies can mitigate such effects, may provide further insights into the integration of missing data methods with modern machine learning workflows.